

Ronny Cravioto-Ross

UBC Engineering Physics Student

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Engineering Physics student with experience writing Rust, C/C++, and Python for normal and bare metal (no_std) targets, experience with robotics, skilled at learning fast, asking the right questions, and building things from the ground up. Conscious of software design principles (e.g. design by contract, encapsulation, etc), yet passionate for rapid prototyping and getting hands dirty.

Experience and Projects

Research Associate – Blue Sky Spectroscopy

(Co-op, On-Site) 2025 Jan. — 2025 Apr.

- Developed multi-channel PID temperature controller through the full design process (PCBs w/ KiCAD, software, testing, etc), for temperature control in large volume cryostats.
- Created and tested PID implementation and I2C ADC driver in C++.
- Reliably controlled temp to within $\pm 3\text{mK}$ of setpoint, 40% better than the target $\pm 5\text{mK}$.
- Worked with vacuum and cryogenic systems, built cryostat test jigs and wire harnesses.
- Worked highly independently, freely choosing and implementing major design components.

Robot Summer (UBC Engineering Physics Competition) - First Place of 15 Teams! – See this video; purple bot is mine: youtu.be/nPeLP5-Fo64

(UBC Course, Project) 2025 May. — 2025 Aug.

- Worked with 3 peers to develop an autonomous robot from scratch, from mechanical to electrical to software, with resilient design as our core philosophy, on a tight 6 week timeline.
- Won 1st place (of 15) thanks to my team's philosophy of fast iteration and ability to pivot
- Wrote ~50% of firmware codebase (C++, 4.5k lines total), using FreeRTOS and implementing PWM/H-bridge based PID motor control, bit-banging shift registers, I2C sensors, interrupt-based quadrature encoder reader, interrupt-based stepper signal wavegen, odometry position tracking, cubic spline inverse trig functions for arm positioning with constrained compute.
- Designed and built all robot electrical systems (incl. H-bridge and motherboard PCBs).
- Created (in a weekend) a graphical serial debug interface in rust, allowing for live PID tuning, odometry visualization, and arm control. UART (with ring buffers for efficiency) was used, and commands/numeric values (including floats) were sent as raw binary. This became an invaluable insightful resource, allowing us to troubleshoot smarter (not harder).

Neural Network from Scratch (Rust) – Implemented a two layer feed-forward NN from scratch to detect handwritten numbers.

(Personal Project) 2025 Feb. — 2025 Feb.

- Used book as guideline for how to construct/train a NN to recognize handwritten numerical digits with the MNIST dataset.
- Implemented multi-threaded stochastic gradient descent with hand coded linear algebra, from scratch (no math/AI/ML deps).
- Created GUI for training and testing with the egui graphics rust crate. After training, user can draw on a canvas, and the NN accurately guesses what the user draws (digits from 0-9) in real time.

Guitar Multi-Effects Pedal with Bare Metal Rust (In Progress)

(Personal Project) 2025 Apr. — Ongoing

- Goal: Create a low-latency guitar effects platform using the RPi Zero 2 W and I2S 24/32 bit 192kHz ADC and DAC.
- Bare-metal rust code has been successfully executed, able to blink LED and output through UART.
- PCB which provides synced clock signal and breakout pins for I2S ADC and DAC has been built and tested.
- Completed: Run Linux on one of 4 cores and execute bare-metal on other three cores, allowing for the convenience of Linux drivers without sacrificing speed to the scheduler; goal is an ultra-fast bare-metal DSP pipeline for near-zero latency.
- Ongoing: Develop communication protocol/driver between Linux userspace and bare-metal effects for real-time parameter tuning (key for 'wah' effect and similar) and signal chain reconfiguration.
- Ongoing: Develop ultra-tight DSP pipeline with the 3 bare metal cores using custom fixed-point representation, dma, cpu mailboxes, cache prefetching, and SIMD assembly optimizations.

Linux Kernel NVIDIA Workaround

(Personal Project) 2024 Aug. — 2024 Aug.

- Tried to enable hybrid graphics for my Linux running (NixOS) laptop, but graphics card refused to sleep.
- Used kernel logs and ftrace to figure out that a faulty ACPI call caused removal of NVIDIA audio device.
- Wrote a kernel patch to block removal of NVIDIA devices by injecting an if statement to 'acpi_hotplug_work_fn.'
- Idle power went from 14W to 8W, significantly improving battery life.

Education

The University of British Columbia - Current Student

B.ASc in Engineering Physics

Minor in Honours Math

Skills/Exposure

Software • Rust • C/C++ • Bare-metal, Embedded • Python • Java • Linux + Commandline • Kernel Compilation • Data logging + analysis •

Electrical: • KiCAD • Circuit Design • 'Scopes, Spectrum Analyzers • Soldering •

Mechanical: • 3D Printing • Vacuum/Cryo • Solidworks/ Fusion360/FreeCAD •